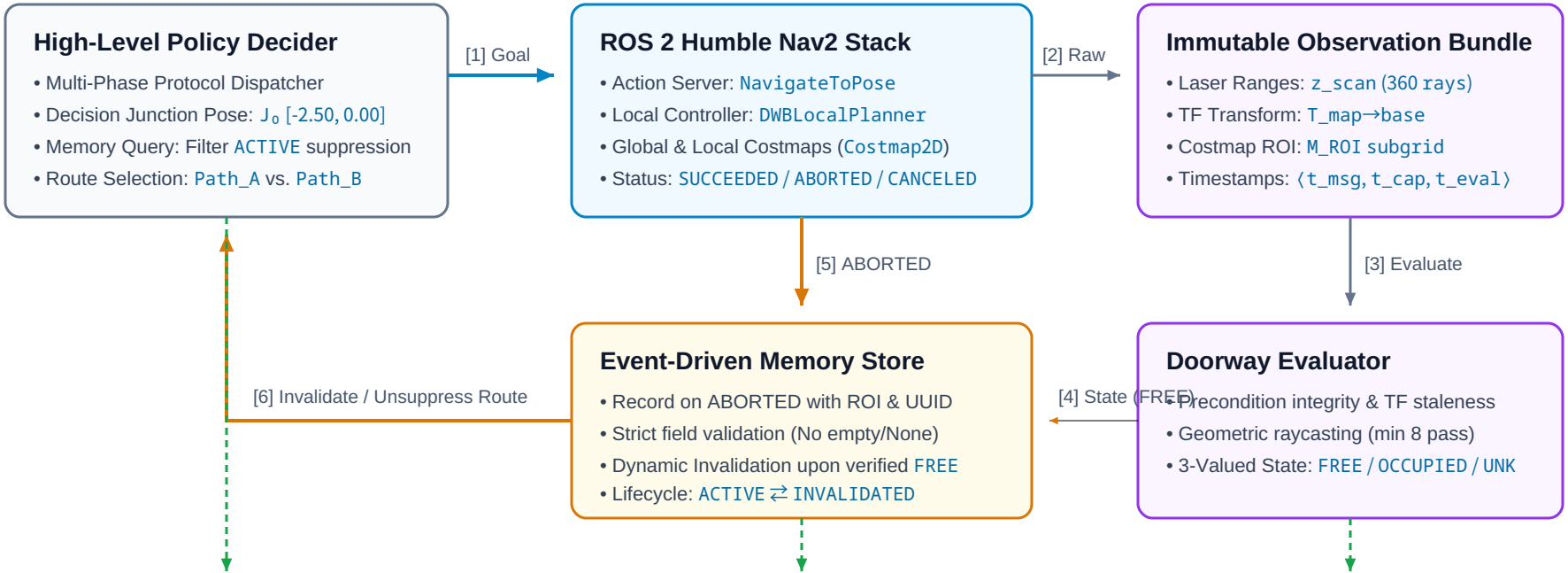
FailMem Architecture & Independent Replay Verification Pipeline

Event-Driven Failure Memory with Immutable Observation Bundles and Physical Replay Audit



Independent Replay & Objective Scoring Auditor (Offline)

- **Anti-Tamper**: Verifies SHA256 hashes against checksums.sha256 manifest
- **Raw Recomputation**: Re-casts rays and subgrids from raw arrays (does not trust online labels)
- **Causality & Physics**: Verifies temporal causality (t_rec ≥ max(t_act, t_obs) - 0.05s) and strict halt stability (|v_lin| ≤ 0.05 m/s, |v_ang| ≤ 0.08 rad/s)